Estrogen Receptor Beta (ERβ) Modulation for Long Term Endometriosis Treatment Design Proposal

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Executive Summary

Endometriosis is a disease of the female reproductive system that has detrimental impacts on approximately **190 million women's physical, mental, and financial health for the duration of their reproductive years**. Current treatment options for endometriosis focus on symptom management, hormone therapy, and highly invasive surgeries. These options are costly, ineffective, and cause patients to experience highly undesirable side effects such as bone density loss, suicidal ideation, vaginal hemorrhaging, and fertility implications. Recent studies have suggested a potential new treatment target for endometriosis: Estrogen Receptor β. **We aim to devise a targeted, non-invasive recombinant DNA therapy for women suffering from endometriosis in order to reduce painful symptoms, minimize hormone-related side effects, and improve long-term reproductive health outcomes.**

In endometrial cells, there are two primary estrogen receptors, ERα and ERβ. While the alpha receptor is associated with healthy endometrial cell growth and estrogen absorption, the beta receptor is found upregulated in endometrial lesions. Furthermore, ERβ enables the accumulation of estrogen and excessive growth of endometrial tissues. With this novel target, our **design proposes the degradation of excessive ERβ receptors with a plasmid that incorporates an ERβ binder and a E3 ubiquitin ligase domain to form an ERβ ubiquibody**. Our plasmid will be delivered via a lipid nanoparticle, and propagated using *E.coli* bacterial transformation. Treatment will subsequently be administered to endometriosis tissue to trigger targeted ubiquitination of ERβ. As alternative designs, we propose to incorporate an inducible promoter using modified estrogen responsive elements as well as investigate the efficacy of MKRN3 overexpression.

Our design will need to meet the following criteria to be considered successful: **effective at targeting ERβ**, **cleared for long term side effects**, **cheap and accessible**, and **safe** as well as **easy to administer.** To ensure safe and effective decrease of the Estrogen Beta Receptor, our testing plan will go through five phases: Design of Peptide Binders, In Vitro testing of Peptide Binders, In Vitro Testing of Ubiquibodies, In Vivo Mice Testing, and Human Clinical Trials. In individuals with endometriosis, this direct targeted reduction of Estrogen Receptor β will lead to a healthy reestablishment of the ERα: ERβ ratio. The re-establishment of a healthy ratio will not disrupt healthy ERα activity thus preventing long term side effects that relate to fertility, reproductive development, healthy bone growth, or side-effects from invasive surgeries.

Figure 1. Graphical abstract of the major aspects of our proposed design

Relevant Background

Endometriosis affects approximately 4-15% females of reproductive age across the globe and can cause chronic pelvic pain, painful menstrual cycles, chronic fatigue, dysmenorrhea, dyspareunia and infertility[1]. This condition is characterized by excessive growth of uterine tissues within the uterus, as well as surrounding the ovaries and fallopian tubes, and can even extend to outside of pelvic organs [2].This excessive tissue growth can contribute to ovarian cyst formation, scar tissue formation and fibrous adhesive tissues which can cause increased pain and infertility [2]. Furthermore, endometriosis has severe impacts on women's quality of life during and after reproductive years. Women diagnosed with endometriosis have reported a 70.2% decrease in physical activity and a 39.5% decrease in professional activities. Additionally, 15% of these women had been diagnosed with depression [1]. Moreover, women with endometriosis can suffer from "cumulative life-course" impairment; this phenomenon occurs due to patients' daily activity choices that compound into effects on family planning, educational attainment, career selection, relationships with loved ones, and general emotional and mental well-being [3].

Aside from painful physical and mental/social implications, endometriosis has critical direct and indirect impacts on worldwide economies and healthcare industries. For patients, endometriosis medical treatment can cost $16,573 annually due to inpatient admissions, emergency room visits, physician visits, and obstetrician/gynecologist visits [4]. In addition, studies have shown that endometriosis can indirectly affect patient finances due to reduction in work productivity and unplanned absences costing them an average of $6,298 [5]. Overall, by observing the pervasive influence on patients' physical, mental, social, and financial well-being, endometriosis clearly has a significant impact on society. Consequently, it is critical to understand how endometriosis can arise, the biological factors that influence endometriosis, and how to improve the quality and variety of treatment options for patients.

There are several biological factors involved in the development of endometriosis, but the main hormonal characteristic is high estrogen production and accumulation. These estrogen pathways are regulated by several key hormones, and their receptors: 17β-Estradiol (E2), receptors ERα (coded by ESR1 gene) and ERβ (coded by ESR2 gene) [6], and gonadotropin hormone (GnRH), along with its receptors [7]. E2 is the strongest form of estrogen in the female reproductive system, and will accumulate in endometriosis tissue [6]. As E2 circulates in the blood it is bound by two different types of receptors ERα and ERβ. In healthy endometrial cells E2 primarily binds to ERα, but in endometriosis lesions, E2 accumulates and ERβ activity is heightened [6]. Based on this upregulation of ERβ activity, the following proposed design plans will incorporate modulating ERβ to prevent accumulation of estradiol and binding of estrogen receptors. The significance of ER*β* in endometriosis is echoed by other researchers who have suggested that it is a critical modulator of endometriosis progression due to the fact that cytoplasmic ER*β* stimulates proinflammatory signaling in ectopic lesions and inhibits apoptosis thus promoting cell proliferation. Additionally, ERβ contributes to progesterone resistance in endometriotic lesions by downregulating progesterone receptors. In turn, progesterone resistance facilitates the growth and survival of endometriosis cells [8]. Consequently, ERβ is an attractive therapeutic target to facilitate effective long-term treatment of endometriosis.

The next major hormone that we will investigate for alternative designs, GnRH, is the primary neural regulator of the entire reproductive system, and it controls estradiol production [9]. GnRH is secreted in two methods, pulse generation and surge generation. Abnormalities in GnRH pulse generation are associated with elevated estrogen production levels and endometrial tissue growth [9]. With this background, our alternative recombinant DNA technology design will aim to control this pulse generation via Makorin Ring Finger Protein 3 (MKRN3), which affects the promoter region of the GnRH gene [7]. Previous work has been done to develop hormone receptor antagonizing, predominantly GnRH antagonists [10], but current technologies targeting estrogen receptor beta have been limited to mouse model testing to observe how ERβ blocking affects endometrial cell proliferation and fertilization [11].

Our design proposes using *Escherichia coli (E.coli)* as a biological factory to develop recombinant DNA based therapies that degrade the overexpressed beta estrogen receptor as well as alternatively decrease GnRH expression. In doing so, our goal is to better regulate Estradiol production and accumulation of E2 in endometrial tissue cells.

Markets/Needs Assessment

Currently there is no known cure for endometriosis, which affects approximately 190 million women worldwide [12]. This prevalence underscores the substantial impact of this condition on women's global health. Moreover, it is a chronic disease which affects both the physical and mental wellbeing of patients due to the extreme pain brought about by the condition.

The global endometriosis market, reflecting the demand for diagnosis and treatment, was valued at $2.8 billion in 2022 and is projected to reach $3.9 billion by 2030, growing at a Compound Annual Growth Rate (CAGR) of 4.2% from 2023 to 2030 [13]. This market growth has been catalyzed by several factors including: increasing awareness among both patients and healthcare providers, improved diagnostic techniques such as high-resolution imaging and laparoscopy, and the rising demand for effective treatments that can manage symptoms and improve quality of life. Additionally, the market is being further propelled by ongoing research into novel therapies, including targeted molecular approaches and personalized medicine strategies. The economic burden of endometriosis is considerable for both patients and the healthcare system, with estimates suggesting annual costs of $69.4 billion in the United States alone, encompassing both direct healthcare costs and indirect costs such as lost productivity [14]. This combination of high prevalence, significant economic impact, and growing market size epitomizes the critical need for innovative and effective treatment approaches in the field of endometriosis management.

Most present treatment regimens for endometriosis primarily focus on symptom management which include hormonal therapies and pain medications. However, these approaches often fall short in providing long-term relief and are associated with significant side effects and limitations. Hormonal therapies, such as combined oral contraceptives and gonadotropin-releasing hormone (GnRH) antagonists, while effective in some cases, can lead to undesirable side effects like bone density loss and menopausal-like symptoms [15]. Furthermore, previous studies have shown that hormonal treatment is only effective for about two-thirds of the women suffering from endometriosis, leaving one third of patients under-served. Finally, surgical interventions can be pursued, including laparoscopy and hysterectomy. However, these are highly invasive and may not prevent recurrence of the condition [16].

Our proposed approach involving targeting the beta estrogen receptor (ERβ) and GnRH receptor represents a novel and potentially transformative treatment strategy. To assess its potential impact and market position, we have conducted a comprehensive SWOT analysis in **Table 1** below to showcase how our approach compares in terms of strengths, weaknesses, opportunities, and threats relative to existing treatment options described later on:

Table 1: SWOT Overview of Proposed Ubiquibody Therapy

Strengths	Weaknesses	Opportunities	Threats
● Non-invasive. ● Targeted, focused approach for ER□ downregulation. ● Transient expression. ● Potential to reduce endometrial lesion growth. ● Adaptability to personalized medicine. ● Less likely to be immunogenic as our delivery method leverages lipid nanoparticles (LNPs) as compared to viral particles.	● Potential cost and accessibility issues as a novel therapy. ● Limited data for long-term safety ● Challenges in scaling production and maintaining efficiency for widespread use ● Possible reliance on specialized healthcare providers	● Expands treatment options available to patients suffering from endometriosis, thus filling a gap. ● Potential expansion for treatment of other conditions characterized by the upregulation of ER□. ● Potential for broader applications beyond endometriosis, including conditions like PCOS, uterine fibroids, or other estrogen-related disorders. ● Partnerships with biotechnology and pharmaceutical companies to accelerate development and distribution.	● Regulatory challenges and lengthy approval pathway. ● Potential high initial cost of development and manufacturing of biologic therapies. ● Hesitancy of patients to adopt new and unfamiliar treatments. ● Adverse public perception or resistance to gene therapy due to concerns with genetic modifications.

Our approach has the potential to address several different limitations that exist with current solutions. By specifically targeting the underlying molecular mechanisms of endometriosis, it may offer more sustained relief and perhaps modify the course of the disease. This is important given that current treatments often provide only temporary relief, with numerous patients experiencing recurrence of symptoms post-discontinuation [17]. Moreover, the specificity of our approach in targeting ERβ and GnRH expression may result in fewer systematic side effects compared to relevant hormonal therapies. This could significantly improve patient quality of life and overall treatment adherence, addressing a major challenge in current endometriosis management [18]. The market opportunity for our therapy is considerable, given the large patient population and limitations of current treatments. Additionally, there is a growing trend towards personalized medicine in endometriosis treatment, which our approach could capitalize on by tailoring the therapy to individual patient profiles [19]. However, it is important to note that as a novel therapy, our approach will undoubtedly face challenges in terms of regulatory approval, clinical trial requirements, and initial costs. Patient and physician education will be crucial to ensure acceptance and adoption of this new modality.

In summary, our innovative approach to endometriosis treatment stands at the forefront of addressing a critical gap in women's health care. The substantial patient population and expanding market underscore the pressing demand for advanced therapeutic options. While our strategy faces the inherent hurdles associated with pioneering medical technologies, its capacity to offer precise, enduring symptom management with reduced adverse effects sets it apart from existing interventions. The path forward involves rigorous scientific investigation, clinical validation, and strategic integration into the healthcare landscape. By successfully navigating these challenges, our therapy has the potential to revolutionize endometriosis management, significantly enhancing the quality of life for millions of women worldwide. This endeavor not only represents a scientific breakthrough but also a promising commercial opportunity in the evolving field of reproductive health.

Design Criteria

There are five main design criteria that make our endometriosis treatment competitive, safe, and reliable in the current market. The first two criteria are to ensure patient safety and limited side effects due to the complex and intricate nature of hormone regulation. The next three criteria focus on meeting current treatment gaps that make our product competitive in the market. With this background in mind, our proposed biotechnology will need to meet the following criteria: **effective at targeting** the proposed receptors, **cleared for long term side effects** that might hinder fertility, alter menopause, or healthy reproductive development, **cheap and accessible** by patients from a variety of socioeconomic backgrounds, and **safe** and **easy to administer**. As shown in **Table 2**, our design criteria are focused on resolving common concerns with current treatment options.

Table 2: Comparison of design criteria in proposed design and in current market options for endometriosis treatment [20], [21], [22], [23]

Design Criteria	**Proposed Primary Design (ERβ Ubiquibody) Goals**	**Orilissa® Elagolix**	**Linzagolix**	**NSAIDS**	**Laparoscopy or Laparotomy**
Effective Targeting	Primary Targeting of Estrogen Receptor Beta	GnRH receptor antagonist	GnRH receptor antagonist	Blocks endometriosis-associated prostaglandin formation	Camera view of removal of endometrial lesions
Long Term Effects	Minimal side effects No decrease in bone mineral density	Bone Loss/ decreased bone mineral density Pregnancy loss Suicidal thoughts Menopausal symptoms	Bone loss/ decreased bone density Vaginal hemorrhage Menopausal symptoms Liver disorders	Stomach ulcers Kidney Failure Increased Hypertension	Infection Bleeding Possible injury to nearby organs
Cost and Accessibility	Post manufacturing stability dosage aim list price less than $800.	$1046.66 per dose (list price)	€1461.00 per dosage	$5-8 per daily dosage	$2500-$7500 per surgery based on severity of procedure
Safety	FDA Approval and EU Approval required	FDA Approved	EU Approval under regulatory review in US	FDA Approved	Minimally/moderately invasive surgical procedure
Frequency and Invasivity of administration	Doctor Administered injections every 4-6 months	Once daily oral pill	Once daily oral pill	400mg every 4-6 hours	Return to normal activity after approximately 2 weeks

Effective Targeting

As discussed in the introduction there are two primary receptors for estradiol, estrogen receptor alpha and estrogen receptor beta, and the alpha receptor is considered

part of a healthy well regulated pathway, but the beta receptor is upregulated in endometriosis lesion tissue [6]. This distinction of receptors is critical to ensuring that our proposed design only targets the effects caused by excessive estradiol in the blood. Furthermore, if our design does not sufficiently degrade ERβ this can lead to significant decreases in ERα gene expression in human ovarian cells [24]. In endometriotic lesions, expression of ERβ is 100 times higher than in healthy tissue[25], and this leads to an imbalance in alpha:beta receptors. This imbalance also causes progesterone resistance, another key cause of endometriosis symptoms and disease progression [24]. A healthy ratio of ERα:ERβ in endometrial cells was found to be 15.5 ± 2.8, but this ratio in endometriomas cells was approximately 65% lower at 5.2 ± 0.9 [26]. Given these ratios it is critical that our design sufficiently degrades ER-beta receptors while maintaining ER-alpha receptors to keep this ratio in a healthy range. Our other design strategy focuses on the GnRH expression controlled by MKRN3 gene, which as discussed previously, is responsible for triggering estrogen secretion. Effective targeting in this case means that we must ensure accurate targeting of the MKRN3 gene in the hypothalamus. The location makes effective targeting here a critical design component since the selective semi-permeable membrane between blood and the brain can hinder penetration [27].

Long-Term Effects

The nature of our design is heavily involved in primarily decreasing the natural receptor, ERβ, and secondarily the expression of GnRH. However, both of these play vital roles in the healthy development and functionality of the body. This means it is critical to ensure long term damage to these receptors is not caused, and that in altering the natural hormonal pathways we do not hinder reproductive development, fertility, or menopause if a patient does not sustain treatment with our design. While the complete function of ERβ is still unknown, studies have shown that methylation of the ERβ promoter region has been frequently found in breast cancer cells and this has led to speculation that ERβ is a potential tumor suppressor gene[28]. This connects back to our understanding of the alpha:beta ratio of estrogen receptors as discussed in the effective targeting. Considering that patients' endometriosis symptoms can alleviate once menopause is reached[29], we would need to ensure that once a patient stops taking our hormone treatment a healthy ratio is able to be achieved. With regard to our secondary design, GnRH expression blockers, current gonadotropin-antagonist medicines are already known to put the body in a menopausal-like state and can even weaken bones[29]. Furthermore, the method of altering GnRH expression is an indirect treatment for overproduction and secretion of estrogen. GnRH triggers more directly the release of Luteinizing Hormone (LH) and Follicle-Stimulating Hormone (FSH) which are essential for healthy human reproductive functions[9]. Currently long-term side effects are a prominent concern for GnRH antagonist, with approximately 80% of patients experiencing some form of long term side effect including, but not limited to: migraines, joint pain, bone density loss, memory loss, hypertension, decrease libido, insomnia, anxiety, nerve pain, weight gain, and hot flashes[30]. With this current gap in treatment, it is critical that our novel design mitigates potential negative long-term effects of treatment.

Frequency and Invasivity of Administration

As demonstrated in Table 2, there are a variety of treatment types for endometriosis that target the different stages of disease progression. There are four stages of endometriosis which are classified by the depth, density, and adhesion of lesions [31]. Based on these different stages, treatment options vary in how often they are administered and how invasive the administration/treatment option is. Given this breadth of options, our proposed design needs to be minimally invasive and minimizes frequency of professional administration, while still being effective for the appropriate disease stage.

Cost and Accessibility

Race, ethnicity and economic disparity has a long history in both the study and treatment of women with endometriosis[32], and given this background it is critical that our design is accessible to women from a variety of socio-economic backgrounds. Additionally, in current treatment options cost is a large hurdle both in diagnosis and then later in treatment; this emphasizes the importance of making a low cost treatment option. Beginning in the 1950's, endometriosis had been characterized as a disease affecting upper-class White women, and this bias is still present in diagnosis rates between White women and Black and Hispanic women [32]. Additionally, disease symptoms/presentation differs across ethnic groups and disease comorbidities are also different [33]. This means that it is critical for our design to be created and tested on a variety of women to ensure treatment needs for the disease does not negatively interact with treatments of comorbidities and environmental factors. As stated in the background, the average cost of annual treatment is $16,537 [4] and can cause patients opportunities for economic growth in their professional life. Taking these high costs into account, drives our desire to have a design that is cheap to manufacture post research and development.

Intellectual Property and Claims

Patent Landscape Overview

Table 3: Patent Landscape Overview Chart

Field: ERβ inhibitor to treat endometriosis
Search Terms: "Estrogen Receptor", "Endometriosis treatment"
Patent Databases: Google Patents, USPTO
Results: Google Patents: ○ "Estrogen Receptor" → 8814 hits ○ "Endometriosis treatment" → 90 hits USPTO: ○ "Estrogen Receptor" → 12214 hits ○ "Endometriosis" → 1456 hits
Key Patents **Patent WO A1**: The patent, assigned to Abbvie Inc., describes the compositions and processes of Elagolix, which binds to GnRH receptors in a competitive manner and leads to dose-dependent inhibition of endometriotic proliferation [34]. **Patent US10682351B2**: The patent from Abbvie Inc. describes the use of Elagolix, the GnRH receptor antagonist, to treat endometriosis. The patent outlines the dosing schedules and explains the combination of Elagolix with other drugs like rifampin and ketoconazole to enhance efficiency at reducing symptoms associated with endometriosis [34]. **Patent WO**: The patent describes the use of Degarelix, a GnRH antagonist, as a treatment of endometriosis, endometriotic ovarian cysts, and recurrent endometriotic lesions following surgery. Degarelix is hypothesized to inhibit secretion of FSH and LH, binding to the GnRH receptors and blocking the interactions [35].

Claims of Novelty and Non-obviousness

Our project introduces a novel treatment for endometriosis by utilizing Lipid Nanoparticles (LNPs) to deliver ubiquibody plasmids that ubiquitinate and regulate the expression of ERβ. This strategy modulates estrogen levels to alleviate symptoms of endometriosis and, unlike any existing patents, has not been documented in previous art.

In contrast to patent WO A1m which delivers the compound as either a solution to the stomach or in an InteliSite® capsule to the jejunum and ileum, our work employs LNPs for injectable delivery to achieve enhanced specificity. Beyond the delivery method, our work regulates levels of ERβ instead of directly suppressing the GnRH receptors to mitigate the risk of abrupt hormone suppression and expectedly to produce fewer side effects and to offer safer and more precise treatment. Lastly, our use of ubiquitination as the regulatory mechanism offers more precise degradation of target, more dynamic and reversible control that facilitates conditional gene expression, and reduced risk of side effects.

In summary, the combination of LNP delivery, ERβ regulation, and ubiquitination-based method introduces a unique and highly specific treatment for endometriosis, which is not addressed by existing patents.

Proposed Design Plan

Since the root causes of endometriosis are influenced by estrogen, estrogen receptors play an integral role in the pathogenesis of this condition. In particular, estrogen receptor beta (ERβ) is significantly elevated in endometriotic tissues compared to the normal endometrium. Bulun et al have suggested that the levels of ERβ are >100x higher in endometriotic tissue compared to the normal endometrium [25].

Our proposed design aims to effectively bind to ERβ in endometriotic tissues and degrade these receptors in an attempt to restore balance between ERα and ERβ levels. In order to achieve this goal, we designed a plasmid which incorporates a novel binder to ERβ linked with E3 ubiquitin ligase domain, forming an ERβ ubiquibody. Targeted ubiquitination of ERβ, specifically in endometriotic tissue may prove to be beneficial as it would facilitate the decrease of ERβ by leveraging the body's natural protein degradation machinery.

To form this ubiquibody we designed a plasmid that incorporates several key components to ensure effective expression, targeting, degradation, and tracking of ERβ in endometriotic tissues. The core of our strategy relies on a peptide binder that specifically targets ERβ. To ensure flexibility and functionality, we have included a GS linker that connects the peptide binder to the E3 ubiquitin ligase domain, CHIPΔTPR. This domain has been shown to efficiently mediate the ubiquitination of target proteins, marking them for degradation by the proteasome [36]. We sourced the sequence for CHIPΔTPR from UniProt and then codon-optimized it for efficient expression in human cells.

In order to ensure robust expression in mammalian systems, we have incorporated a CMV promoter, which will drive high-level transcription of our ubiquibody construct. The CMV promoter has been proven to be effective in overexpressing genes in endometrial cells [37]. We also included a Kozak sequence upstream of the start codon to enhance translation initiation in eukaryotic cells, ensuring that the ubiquibody is produced at high levels. Additionally, we included a T7 promoter to allow flexibility for bacterial or in vitro expression, making the plasmid adaptable for different experimental setups. To ensure efficient mRNA processing, we added a polyadenylation (poly A) terminator, which enables the export and stability of the mRNA transcript.

An ampicillin resistance gene was included to allow for selection of plasmid-containing *E. coli* cells during cloning, which allows for easy propagation in bacterial cultures. To track the expression and localization of the ubiquibody in live cells, we linked it to mCherry, a fluorescent reporter, allowing us to visually confirm expression and intracellular distribution. We also incorporated an IRES2 sequence, enabling the co-expression of both the ubiquibody and the mCherry reporter from a single mRNA

transcript, ensuring that both components are produced efficiently. Moreover, the inclusion of a HIS tag was executed to facilitate easy protein extraction and purification.

Figure 2: Plasmid map of proposed plasmid construct containing our ubiquibody. Vector backbone adapted from BME305L (Molecular Engineering).

The gene fragments encoding the designed peptide binders, linker and E3 ubiquitin ligase domain can be purchased online via platforms such as TWIST bioscience or IDT and plasmids will be cloned using Gibson Assembly.

Testing Plan

Our testing plan will be divided into **5** key phases. In ***Phase 1*** of our testing plan we will create and screen peptide binders targeted to ERβ. In order to effectively target ER*β,* a novel peptide binder is essential. Specific peptide binders will be designed using RoseTTAFold Diffusion(RFDiffusion) and PepMLM. Multiple candidate binders will be created via these methods and subsequently tested in-silico using AlphaFold3. In silico testing will provide valuable insight into how likely our proposed binders will bind to ER*β* prior to in-vitro testing. The most viable protein binder candidates will be selected for in-vitro testing. *(see Appendix 1 for library of potential binders)*

In ***Phase 2***, we will select the top 3 candidate peptide binders to conduct in-vitro testing. The binders will be cloned in the plasmid shown below in **Figure 2** via Gibson Assembly. The vector backbone was obtained from Addgene Plasmid #60733 [38]. The plasmid encodes the peptide binder along with a 6x His tag, which enables purification via affinity chromatography. To drive strong expression, it incorporates the T7 promoter and the efficient ribosome binding site from bacteriophage T7 gene 10, ensuring high levels of protein production when induced with IPTG. The T7 terminator ensures proper transcription termination and stability of the mRNA. These plasmids will then be transformed in *E.Coli* and

plated on kanamycin plates to facilitate amplification. Successfully transformed *E.Coli* colonies will be selected using a kanamycin resistant gene and the sequence will be verified via Sanger Sequencing.

Figure 3. Plasmid map of construct to produce peptide binder

Next, isolation and purification of the correct plasmids will be conducted using a miniprep. The plasmids will then be used to express the peptide binders in *E. coli* by inducing protein production with IPTG. After expression, the peptides will be extracted using immobilized metal affinity chromatography (IMAC) to purify the His-tagged proteins. The purified peptides will be quantified using a Bicinchoninic Acid (BCA) assay to determine concentration. Once we have the purified peptides, their affinity for ERβ will be tested using a binding assay such as Surface Plasmon Resonance (SPR) or modified ELISA. We will primarily use SPR to analyze metrics such as the dissociation constant (Kd) to determine the binding affinity of the peptides for ERβ. In this binding assay, we will also investigate the affinity of the peptides for ERα to verify the specificity of our binders solely to ERβ.

In ***Phase 3***, once the most suitable binder has been selected after in-silico and in-vitro testing, this binder will be linked to an E3 ubiquitin ligase to facilitate degradation of ER*β* upon interaction with the binder. A similar plasmid cloning, isolation and purification process will be conducted as detailed in ***Phase 2***. The only difference in this phase is the inclusion of a linker and E3 ubiquitin ligase along with the protein binder as our DNA insert. The degradation ability of our ubiquibody will be tested in-vitro. The in-vitro testing plan will involve transfecting an endometriotic cell line which naturally overexpresses ERβ with our plasmid containing the ubiquibody and subsequently quantifying ubiquitination via a Western Blot. A Western Blot will enable us to measure the degradation of ERβ by analyzing the overall levels of protein. As a negative control we will also transfect endometriotic cells with plasmids which do not contain the ubiquibody.

In ***Phase 4***, in-vivo studies will be conducted using mice induced with endometriosis by injecting endometrial tissue into the peritoneal cavity [39]. In this phase, the ubiquibody plasmid will be encapsulated in lipid nanoparticles (LNPs) for delivery. The LNPs will be prepared, and their size and encapsulation efficiency will be validated. Once the

LNP-encapsulated ubiquibody plasmid is ready, it will be delivered via localized injection into the peritoneal cavity, following a scheduled treatment regimen. Upon treatment, the size and number of endometriotic lesions will be monitored before and after each treatment session. Fluorescently tagged LNPs or the reporter gene (mCherry) within the plasmid will be used to track the distribution and uptake of the LNPs in target tissues. Additionally, the activity and decrease of ERβ in the mice can be tracked using IVIS imaging to observe the impact of the treatment on ERβ expression in real time. This approach will allow for continuous monitoring of the therapeutic effect, as well as assessment of the reduction in endometriotic lesion size and ERβ activity over the course of the treatment.

In ***Phase 5***, human clinical trials will be conducted. Based on the efficacy of our design in the in-vivo mouse model, we will further refine our ubiquibody. For delivery in human patients we will once again leverage lipid nanoparticles that are specifically targeted to ERβ, through the use of active targeting, where a ligand is incorporated on the surface of the nanoparticle to facilitate binding to the ERβ. Further, to ensure that we are targeting ERβ in endometriosis cells, the LNPs containing the plasmid will be delivered via an intraperitoneal injection.

Throughout the clinical trial, the health of patients will be consistently monitored to ensure unwanted or off-target effects are not experienced. Ultimately, the key metrics investigated will be the size, number and presence of endometriotic lesions (measured through imaging) as well as changes in pain experienced (measured using the Numeric Rating Scale) and the overall quality of life.

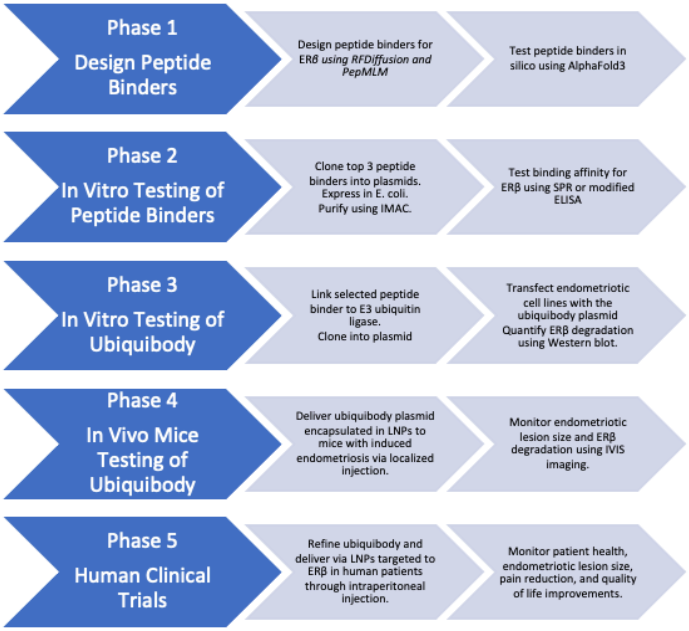


Figure 4. Flowchart detailing testing plan for ubiquibody design therapy

Alternative Designs

One of the major challenges with our design is the need for precise regulation of estrogen receptor beta activity in order to maintain normal physiological functioning. In healthy tissues, the balance between ERα and ERβ is heavily regulated as both receptors are essential for mediating estrogen's effects on reproductive, cardiovascular, skeletal, and central nervous systems [40]. Disrupting this balance by continuously degrading ERβ for the lifespan of the LNP, as our ubiquibody design intends, could lead to unwanted side effects, such as impaired fertility or cardiovascular issues [40]. To address this we propose an alternative design that could involve using an estradiol-inducible promoter [41]. This promoter would activate the expression of the ubiquibody only when estradiol levels are significantly elevated. High estradiol levels are characteristic of endometriotic tissues. Therefore, by linking the ubiquibody expression to the local hormonal environment we could

ensure that ERβ degradation is dynamically regulated.

Another potential approach to managing endometriosis would be targeting the underlying hormonal drivers of the disease. This could be done by modulating GnRH pulse generation using Makorin Ring Finger Protein 3 (MKRN3). MKRN3 is a natural repressor of GnRH secretion from the hypothalamus and thus increasing its expression can be used to directly regulate and normalize the pulsatile release of GnRH [41]. This would be achieved by delivering an MKRN3 gene with an AAV to the hypothalamus to overexpress it. By increasing MKRN3 expression we can suppress the excessive GnRH pulses that drive overproduction of luteinizing hormone (LH) and follicle-stimulating hormone (FSH). This in turn controls estrogen production in the ovaries. Therefore, by restoring to normal hormonal rhythms we could reduce endometrial lesions without disrupting other essential reproductive functions.

Stakeholder Analysis and Ethical Considerations

Our project introduces a novel treatment for endometriosis that combines LNP delivery, upstream hormone regulation, inducible gene expression, and ubiquitination-based regulation methods. The following analysis will explore stakeholders' influence, expectations, and the strategies needed to engage them throughout the drug development and product commercialization process. By mapping the stakeholders, we aim to ensure that the treatment is safe, effective, and widely accessible to those in need. Below is a ranked list of the identified stakeholders, ordered by their level of influence.

Relevant Stakeholders

1. Patients with Endometriosis: As the primary beneficiaries of this therapy, patients (primarily women of reproductive age) have the most significant stake in its success. Their experiences, needs, and concerns should be central to the development process. They have high interest but potentially limited direct influence on the development of the therapy. Their feedback on the efficacy of the therapy, side effects, and overall experience will directly influence the success and market adoption of the treatment.

2. Regulatory Agencies: Organizations like the Food and Drug Administration (FDA) in the United States and the European Medicines Agency (EMA) in Europe will have critical influence over the approval and regulation of our proposed therapy. They have the highest influence as they have the decision-making power of whether our drug can be legally developed, examined, and commercialized. Without their approval, the process cannot move forward, regardless of other stakeholders' interests. Their stringent requirements will shape the development and testing processes. They have high influence but neutral interest. The main role of these regulatory industries is to ensure that the therapy meets stringent safety and efficacy standards before it can be distributed publicly. We should keep the stakeholders updated with progress in the project and establish frequent communication plans to facilitate the development process. A comprehensive clinical trial plan should also be negotiated to meet the concern about safety and efficacy from regulatory bodies.

3. Healthcare Providers: Gynecologists, endocrinologists, and other medical professionals treating endometriosis patients will play a crucial role in adopting and administering the therapy. Their expertise and willingness to embrace new treatments will significantly impact the therapy's success. They have both high influence and high interest. Additionally, they serve as gatekeepers who can either facilitate or hinder the use of novel treatments based on the professional treatment.

4. Research Institutions and Biotechnology Companies: These entities, including our own, drive the development of the therapy. They have high interest and moderate to high influence, depending on their involvement in the project. The companies involved in developing and producing the therapy, including those managing the engineering *E.coli* delivery system, have substantial power over the direction of the therapy's commercialization and its affordability. Decisions about pricing, production, and marketing strategies will significantly impact accessibility and market success.

5. Health Insurance Companies: Insurers will play a crucial role in determining the

accessibility of the therapy through coverage decisions. Their policies can significantly impact patient access and the therapy's market success. They have high influence and moderate interest.
6. Patient Advocacy Groups and Non-Government Organizations (NGO): Organizations focused on endometriosis awareness and support can influence public opinion, policy, and research priorities. They have moderate influence and high interest. These organizations can advocate for broader access to the therapy, particularly for underrepresented groups and those in low-resource settings.
7. Ethical Review Boards: These committees will oversee the ethical aspects of clinical trials and therapy development. They have high influence in the approval process and high interest in ensuring ethical standards are met.
8. Investors and Shareholders: Investors provide the capital necessary for research, development, and commercialization. They have influence over the project as they control the financial lifeline. They can influence the direction of the company's efforts, timelines, and even pricing strategies. We will keep the investors updated with moderately frequent communication plans to ensure the funding of the project.

Ethical Considerations

The development and deployment of our novel therapy will adhere to the four pillars of bioengineering ethics namely: principle of autonomy, principle of justice, principle of nonmaleficence, and principle of beneficence.
1. Safety and Long-term Effects: As a novel therapy, there are concerns about potential unforeseen long-term effects. Rigorous testing and long-term follow-up studies will be crucial to address these concerns [42].
2. Equitable Access: Ensuring that the therapy is accessible to all patients, regardless of socioeconomic status, is an important ethical consideration. This includes addressing potential high costs associated with novel therapies [43]. Therapies such as these tend to be expensive due to the complexity of their development and production. It is vital to ensure that this therapy is accessible to the broad patient population. Any form of price discrimination that limits access based on socioeconomic status would be unethical, exacerbating existing healthcare inequalities.
3. Informed Consent: Given the complexity of the therapy, ensuring that patients fully understand the treatment, its potential risks, and benefits is crucial. This may require developing comprehensive education programs for both patients and healthcare providers [44]. The Informed consent process must be rigorous and transparent, giving patients the autonomy to make decisions about their own healthcare.
4. Reproductive Implications: As the therapy targets reproductive organs, potential impacts on fertility and future pregnancies must be carefully studied and communicated to patients [45]. By potentially altering reproductive hormones, this therapy could influence fertility in ways that have yet to be understood. This raises ethical concerns about the reproductive autonomy of individuals undergoing the therapy; thus, it is essential to communicate clearly whether the therapy will have any impact on fertility, family planning, or long-term reproductive health.
5. Data Privacy and Genetic Information: The use of genetic information in developing personalized treatments raises concerns about data privacy and potential misuse of genetic data [46].
6. Environmental Impact: Consideration must be given to the potential environmental impact of genetically modified organisms used in the therapy, including containment and disposal protocols [47]. Rigorous containment strategies must be implemented to prevent unintended environmental release or cross-contamination. Moreover, there may be concerns about how these engineered organisms interact with the human microbiome, and these effects must be studied carefully.

In conclusion, while our approach holds great promise for improving endometriosis treatment, it is crucial to navigate these ethical considerations carefully. Engaging with all

stakeholders, particularly patients and healthcare providers, throughout the development process will be essential. Transparent communication, rigorous safety protocols, and ongoing ethical review will be key to addressing these challenges and ensuring the responsible development and implementation of this innovative regimen.

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Appendix:

A1) Potential 15 amino acid peptide binders for ERβ generated using PepMLM.

Binder	Pseudo Perplexity
KRHKHHILLKSTKSS	8.10170
KRHHHHHILLDTSSK	8.67570
TRHKHLLILKSSSKT	9.95260
KKKHHILLLKDTSSS	6.370900
KKKGILILLKDTSST	12.26000
KKKHIILILLKSSKK	11.62000
TTRHHLHLLSKTSTK	9.99070
KKHKHHLRLLKDSKT	7.254700
TKKGILIIRKKTKST	23.84800
KRKKLHLLRKDDSST	13.62300
KRHHHHIRRKSTSKK	12.87400
SRHHLHHLLLDSSSS	6.90090
KRHHHILLLLDTSST	10.13500
KTRKHILRKLDTSSS	8.058
KKRHHIILRLDTSST	12.34700
KKRKLHLLLLSSSSK	7.46160